Predicting EU27 Greenhouse-Gas Emissions from Sector-Level Data "Given the sector-level GHG emissions of EU27 countries in year t, predict the total EU27 emissions in year t+1"

Group 5

MECA-H419 — Data-Driven Engineering, ULB

June 2026

Abstract

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3.1 Data preparation

The raw EDGAR booklet contains 4,849 rows, each row describing the annual emissions (Mt CO_2-eq) of a (country, sector, substance) triplet over the period 1970–2024. Before any modelling, the data are aggregated to the EU27 level and transformed so that the eight sectoral features are numerically comparable.

3.1.1 Aggregation to EU27 sector level

The EDGAR dataset already provides a pre-computed `EU27` entity (the 27 current EU member states aggregated according to the Union's official composition), which we adopt to avoid re-implementing the aggregation ourselves. For each of the four substances, we sum the contributions of the eight emission sectors — Agriculture, Buildings, Fuel Exploitation, Industrial Combustion, Power Industry, Processes, Transport, Waste — yielding a clean matrix

$$E \in \mathbb{R}^{T \times S}, \qquad T = 55,\ S = 8,$$

where $E_{t,s}$ is the total CO_2-equivalent emission of sector s in year t. After aggregation, E contains no missing values, which is a key argument for working at the EU27 level rather than at the country level (where several time series are incomplete, especially for the pre-1990 period). The target variable is the annual total

$$y_t \;=\; \sum_{s=1}^{S} E_{t,s}, \tag{1}$$

which drops from $4572\,\mathrm{Mt\,CO_2{-}eq}$ in 1970 to $3165\,\mathrm{Mt\,CO_2{-}eq}$ in 2024 (-30.8%, see Figure 1).

3.1.2 Log-transformation

Raw sectoral emissions span almost an order of magnitude — the Power Industry alone emits over $1300\,\mathrm{Mt\,CO_2{-}eq/yr}$ in the late 1980s, whereas Waste stays below $200\,\mathrm{Mt\,CO_2{-}eq/yr}$ for the entire period. Such heterogeneity makes a regularised regression unfair: the penalty term

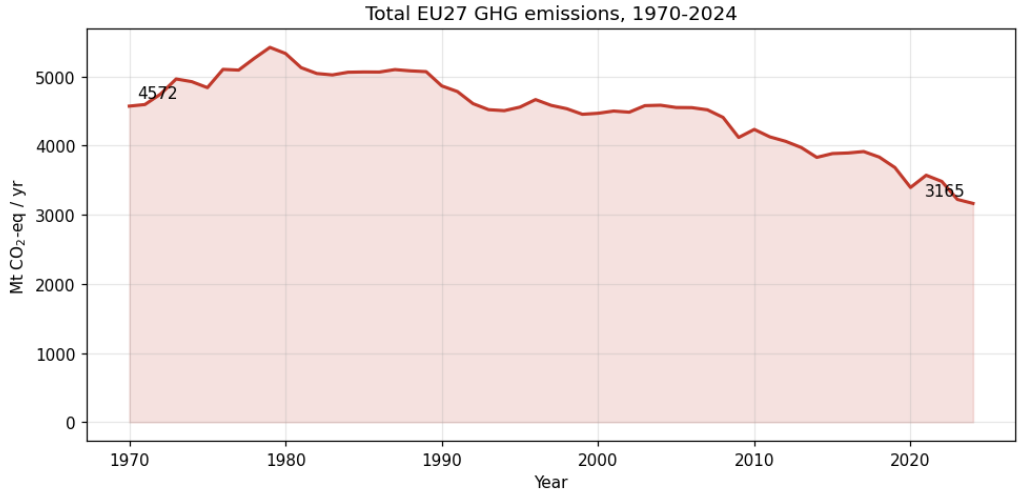

Figure 1: Total EU27 greenhouse-gas emissions, 1970–2024. Aggregated from EDGAR 2025 across the eight sectors and four substances.

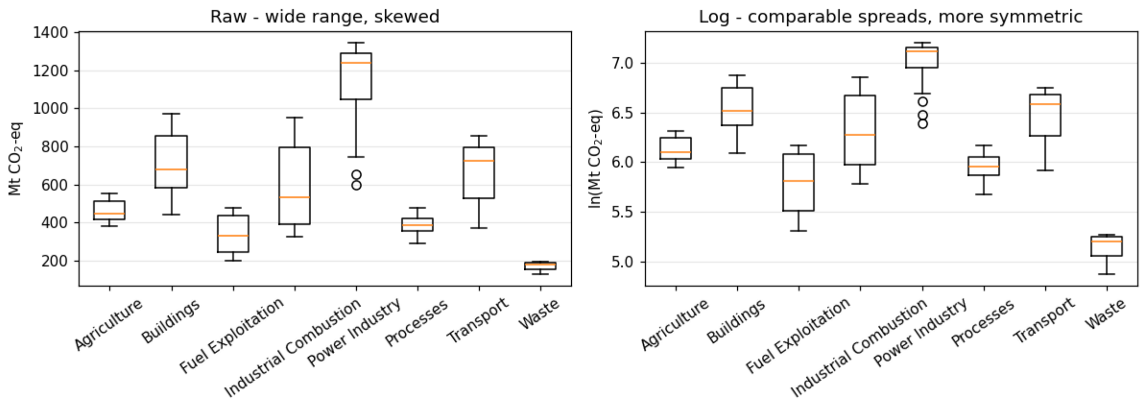

Figure 2: Distribution of sectoral emissions before (left) and after (right) the log-transformation. The log fixes the right-skew and brings sectors onto comparable spreads.

$\lambda\|\beta\|$ would mechanically shrink the small-magnitude features more aggressively. Because all entries of E are strictly positive, we apply a natural logarithm element-wise,

$$\tilde{E}_{t,s} \;=\; \ln E_{t,s}, \qquad \tilde{y}_t \;=\; \ln y_t, \tag{2}$$

which compresses the dynamic range from a factor of ~ 7 to a factor of ~ 1.5 and stabilises the variance (Figure 2). It also linearises any underlying exponential growth or decay, which makes the resulting linear regression interpretable as a model of relative (percentage) changes.

3.1.3 Standardization

After the log-transform, the eight features still have different means ($\overline{\tilde{E}_s}$ ranges from ~ 5.1 for Waste to ~ 7.0 for Power Industry). To put them on the same scale, we apply a z-score normalisation

$$z_{t,s} \;=\; \frac{\tilde{E}_{t,s} - \mu_s}{\sigma_s}, \qquad \mu_s = \frac{1}{T}\sum_t \tilde{E}_{t,s}, \quad \sigma_s^2 = \frac{1}{T}\sum_t \big(\tilde{E}_{t,s} - \mu_s\big)^2. \tag{3}$$

After standardisation each feature has zero mean and unit variance, so Ridge and Lasso penalties act symmetrically on every sector (Figure 3, right panel).

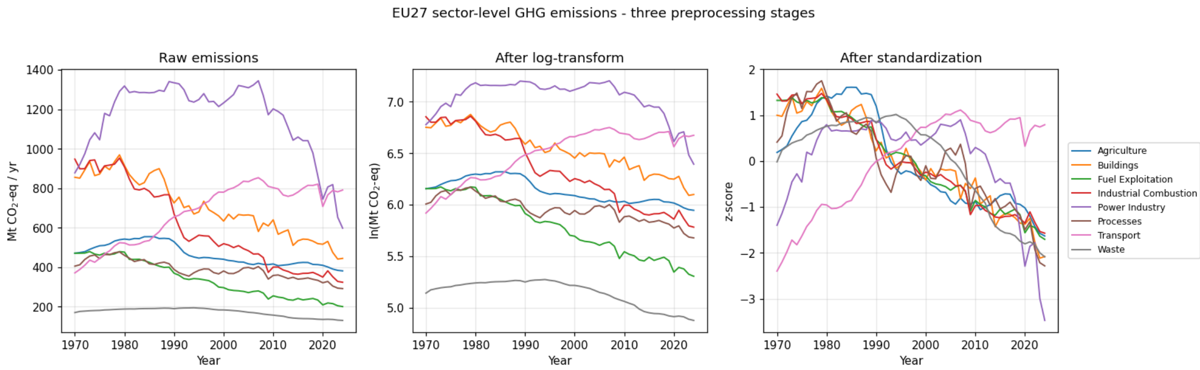

Figure 3: Sectoral EU27 emissions at the three stages of preprocessing: raw (left), after natural logarithm (centre), and after z-score standardisation (right). The transformations preserve the temporal patterns while equalising the numerical scale of the eight features.

A note on data leakage. For the final modelling, μ_s and σ_s are recomputed from the *training fold only* and then applied unchanged to the validation/test folds; otherwise the test statistics would influence the scaling. The global statistics shown in Figure 3 are used only for exploratory visualisation.

3.1.4 Supervised learning setup

To address the research question *"given the sector-level emissions in year t, predict the total emissions in year $t+1$"*, we build a one-step-ahead supervised dataset. Each predictor row is a vector of the eight log-transformed sector emissions, and the corresponding target is the next-year log-total:

$$\mathbf{x}_t = (\tilde{E}_{t,1}, \ldots, \tilde{E}_{t,S}) \in \mathbb{R}^8, \qquad y'_t = \tilde{y}_{t+1} \in \mathbb{R}, \qquad t = 1970, \ldots, 2023. \tag{4}$$

This yields $N = 54$ paired observations $\{(\mathbf{x}_t, y'_t)\}_{t=1970}^{2023}$, with predictors covering 1970–2023 and targets covering 1971–2024. Table 1 summarises the pipeline.

Table 1: Summary of the preprocessing pipeline for the real-world dataset.

Stage	Operation	Resulting shape
Raw EDGAR booklet	Filter `EU27`, $4\times8-7$ rows	25×55
Aggregation	Sum substances per sector (Equation (1))	$E \in \mathbb{R}^{55\times8}$
Log-transform	Natural log of E and total (Equation (2))	$\tilde{E}$, $\tilde{y}$
Standardization	z-score on training fold (Equation (3))	z, $\tilde{y}_{\text{std}}$
Supervised pairs	$(\mathbf{x}_t, y'_t) = (\tilde{E}_t, \tilde{y}_{t+1})$ (Equation (4))	$N = 54$

3.2 Clustering of countries

3.3 Predicting EU27 emissions

3.4 Forecast 2025–2034

4 Discussion and Conclusions

AI Disclosure

We use Claude, ChatGPT, GitHub Copilot, etc for these tasks ...

References

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A Appendix